



Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Group and Ring Theory
MATP33
Wednesday, 30 October 2024
Duration: 08:00–13:00

No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. Please write your anonymous identification number on each sheet of paper.

1. Suppose that e is an idempotent in a ring R with unity, thus $e^2 = e$. Show that as a left R -module $R = Re \oplus R(1 - e)$.
2. Determine the Smith normal form and the rank of the integer matrix

$$\begin{pmatrix} 2 & 3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 10 \end{pmatrix}.$$

3. Find the number of non-isomorphic abelian groups of order 99. Does there exist a non-abelian group of order 99?
4. Let $A \in M_n(\mathbb{C})$ be a nilpotent complex matrix, thus $A^k = 0$ for some positive k . Show that $A^n = 0$ as well.
5. Decompose into the direct sum of cyclic groups: a) $\mathbb{Z}^2 \otimes_{\mathbb{Z}} \mathbb{Z}^6$, b) $\mathbb{Z}_2 \otimes_{\mathbb{Z}} \mathbb{Z}_6$.
6. Let R be a left Artinian ring with unity that is not a division ring. Show that R contains two non-zero elements such that their product is zero.